A growing family of exact reduced-pair cycles

For every $t \geq 0$, Lean checks $f = X^n + 1$ over F_3, $g = 12t + 2$ and $n = 24t + 5$. The actual polynomial is monic, separable and has degree $2g + 1$. The complete finite universe of valid reduced pairs contains n distinct smooth pairs: identity and $(X^d, \pm 1)$, $1 \leq d \leq g$. Normalized addition by $(X, 1)$ cycles through them with exact period n. [pcc:unverified]

The valid pair $(X^2 + X + 2, X)$ is nonsmooth for every t and outside that cycle. Thus the cycle smoothness proportion is 1 and the ambient proportion is strictly smaller. Genus and period are unbounded. [pcc:unverified]

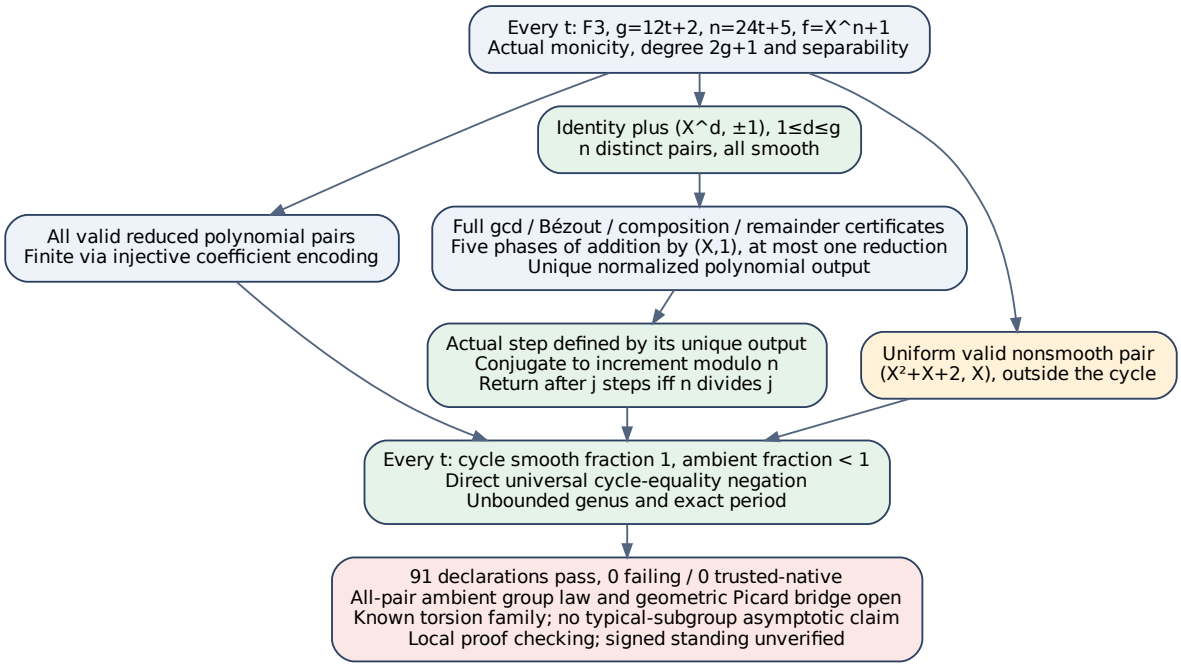


Figure 1: Actual polynomial objects, exact arithmetic and counting. [pcc:unverified]

What is proved, and what remains

The ambient set consists of all monic u with $\deg(u) \le g$, all v with $\deg(v) < \deg(u)$, and u dividing $f - v^2$. Its finiteness is proved by an injective bounded coefficient encoding. The n cycle pairs are proved distinct. The quadratic X^2+X+2 has no F_3 root. Explicit quotient identities, its period dividing X^8-1 and $n=8(3t)+5$ prove the uniform nonsplit pair valid. No finite list is declared complete by assumption. [pcc:unverified]

Every addition step has full monic gcd/Bézout, composition, bounded remainder and zero-or-one reduction certificates. These cover identity, positive increment, the degree-g crossing, negative decrement and cancellation. The actual step is defined from the unique normalized output; agreement with an indexed successor is then proved. Injective positions modulo n prove return after j steps iff n divides j, including the absence of every shorter positive period. [pcc:unverified]

Module	Main obligation
ReducedPairs / CurveFamily	Actual finite universe, monicity/separability/degree, valid split and nonsplit pairs
OrbitFamily	n distinct smooth pairs, fraction 1 versus ambient fraction <1
NormalizedCantor / FamilyTransitions	Unique normalized output and all five polynomial run certificates
Successor / Cycle	Independently defined arithmetic step; exact period n
FamilyCertificate	Unbounded genus/period, combined certificate and direct universal cycle-equality negation

The whole project audit passes 91 declarations, zero failures and zero trusted-native exceptions. Lean 4.35.0 / Mathlib 56a00585; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, linter waivers or added elaboration settings. All 39 attempt/source files and five gate files have byte-equal archive roundtrip checks. Source commit `9e23de04` precedes `PCC-20261110-156a3c`; full local verification passes at 13:09:19Z. Signed standing remains unverified; see verification.json. [pcc:unverified]

Stoll 2014, Lemmas 4.16–4.17, Theorem 4.18 and Example 4.19, printed pages 23–26, were reread. They motivate the actual reduced-pair conditions and addition equations. The geometric Picard/Riemann–Roch correspondence and a full all-pair ambient group law remain separate Lean obligations. This cycle theorem is not promoted to an unconditional geometric Jacobian subgroup family. [pcc:unverified]

Jędrzejak 2016 (DOI 10.4064/aa8141-3-2016), relevant introduction, theorem/model and good-reduction sections and the odd-degree torsion proof on printed page 111 were read, not the full paper. The torsion family is known; no novelty claim. Cantor's 1987 original remains unread after AMS HTTP403. Exact source URLs, hashes and read scopes are in source-review.json. No claim against Gaudry's fixed-genus $q \to \infty$ estimate or typical/large-prime subgroup asymptotics. Periods n are not all prime. [pcc:unverified]

The existing IDC1hb7102c15384f2648 is copied byte-identically for its base-quintic scalar identity only. New characteristic-3 and variable-genus arithmetic rests on Lean. No fresh sampling, native experiment, timing or paid compute. KB/Conductor unavailable; primary/repository/Mathlib fallback. See README.md, sources.json and verification.json. [pcc:unverified]